

Ex. Part A1: Write C program to find the roots of quadratic equation using else if ladder.

Source Code:

```
#include <stdio.h>

#include <math.h>

void main()
{
    float a, b, c;

    float discriminant, root1, root2, realPart, imagPart;

    printf("Enter coefficients a, b and c: ");

    scanf("%f %f %f", &a, &b, &c);

    discriminant = b * b - 4 * a * c;

    if (discriminant > 0)
    {
        root1 = (-b + sqrt(discriminant)) / (2 * a);
        root2 = (-b - sqrt(discriminant)) / (2 * a);
        printf("Roots are real and different.\n");
        printf("Root 1 = %.2f\n", root1);
        printf("Root 2 = %.2f\n", root2);
    }
    else if (discriminant == 0) {
        root1 = -b / (2 * a);
        printf("Roots are real and equal.\n");
        printf("Root = %.2f\n", root1);
    }
}
```

```

else {

    realPart = -b / (2 * a);

    imagPart = sqrt(-discriminant) / (2 * a);

    printf("Roots are complex and imaginary.\n");

    printf("Root 1 = %.2f + %.2fi\n", realPart, imagPart);

    printf("Root 2 = %.2f - %.2fi\n", realPart, imagPart);

}

}

```

Output:

```

Enter the coefficients a, b and c: 1 -3 2
Roots are real and different
Root 1: 2.00
Root 2: 1.00

```

```

Enter the coefficients a, b and c: 1 -2 1
Roots are real and equal
Root 1: 1.00

```

```

Enter the coefficients a, b and c: 1 2 5
Roots are complex and imaginery
Root 1: -1.00 + 2.00
Root 2: -1.00 - 2.00

```

Ex. Part A2: Write C program to read two integer values and an operator has a character and perform basic arithmetic operation on them using switch case (+, -, *, /).

Source Code:

```
#include <stdio.h>

main()
{
    int num1, num2;
    char operator;
    int result;
    printf("Enter first integer: ");
    scanf("%d", &num1);
    printf("Enter second integer: ");
    scanf("%d", &num2);
    printf("Enter an operator (+, -, *, /, %%): ");
    scanf(" %c", &operator);
    switch (operator) {
        case '+':
            result = num1 + num2;
            printf("%d %c %d = %d\n", num1, operator, num2, result);
            break;
        case '-':
            result = num1 - num2;
            printf("%d %c %d = %d\n", num1, operator, num2, result);
            break;
```

```

case '*':

    result = num1 * num2;

    printf("%d %c %d = %d\n", num1, operator, num2, result);

    break;

case '/':

    if (num2 != 0) {

        result = num1 / num2;

        printf("%d %c %d = %d\n", num1, operator, num2, result);

    } else {

        printf("Error: Division by zero is not allowed.\n");

    }

    break;

default:

    printf("Error: Invalid operator.\n");

    break;

}

}

```

Output:

```

Enter first integer: 100
Enter second integer: 50
Enter an operator (+, -, *, /): +
100 + 50 = 150

```

```

Enter first integer: 100
Enter second integer: 50
Enter an operator (+, -, *, /): *
100 * 50 = 5000

```

```

Enter first integer: 100
Enter second integer: 50
Enter an operator (+, -, *, /): -
100 - 50 = 50

```

Ex. Part A3: Write C program to reverse a number to check it is a valid palindrome and to calculate the sum of digits.

Source Code:

```
#include <stdio.h>

main() {

    int num, originalNum, reversedNum = 0, digit, sumOfDigits = 0;

    printf("Enter an integer: ");

    scanf("%d", &num);

    originalNum = num;

    while (num != 0){

        digit = num % 10;

        reversedNum = reversedNum * 10 + digit;

        sumOfDigits += digit;

        num /= 10;

    }

    if (originalNum == reversedNum)

        printf("%d is a palindrome.\n", originalNum);

    else

        printf("%d is not a palindrome.\n", originalNum);

    printf("Reversed number: %d\n", reversedNum);

    printf("Sum of digits: %d\n", sumOfDigits);

}
```

Output:

```
Enter an integer: 123
123 is not a palindrome
Reversed number: 321
Sum of digits: 6
```

```
Enter an integer: 232
232 is a palindrome
Reversed number: 232
Sum of digits: 7
```

Ex. Part A4: Write C program to generate N Fibonacci number.

Source Code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int f1=0,f2=1,f3,i,n;
    clrscr();
    printf("Enter the value for N: ");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        printf("%d\t",f1);
        f3=f1+f2;
        f1=f2;
        f2=f3;
    }
    getch();
}
```

Output:

```
Enter the value for N: 10
0      1      1      2      3      5      8      13      21      34
```

Ex. Part A5: Write C program to find the given number is a prime number or not.

Source Code:

```
#include<stdio.h>

void main()
{
    int n,i,flag=0;
    clrscr();
    printf("Enter a positive integer : ");
    scanf("%d",&n);
    for(i=2;i<=n/2;++i)
    {
        if(n%i==0)
        {
            flag=1;
            break;
        }
    }
    if(flag==0)
        printf("%d is a prime number",n);
    else
        printf("%d is not a prime number",n);
    getch();
}
```

Output:

```
Enter a positive integer: 2  
2 is a prime number
```

```
Enter a positive integer: 22  
22 is not a prime number
```

Ex. Part A6: Write C program to count occurrences of each character in a given string.

Source Code:

```
#include<stdio.h>

#include<conio.h>

int main()
{
    char str[20],ch;
    int count=0,i;
    clrscr();
    printf("Enter a string: ");
    scanf("%s",str);
    printf("Enter the character to be searched: ");
    scanf(" %c",&ch);
    for(i=0;str[i]!='\0';i++)
    {
        if(str[i]==ch)
            count++;
    }
    if(count==0)
        printf("Character '%c' is not present",ch);
    else
        printf("Occurence of character'%c':%d",ch,count);
    getch();
}
```

Output:

```
Enter a string: Hello123#  
Enter the character to be searched: l  
Occurence of character 'l':2
```

```
Enter a string: Hello123#  
Enter the character to be searched: a  
Character 'a' is not present
```

Ex. Part A7: Write C program to read with alphabets, digits, and special characters and convert letters into uppercase and lowercase and vice-versa and retain the digits and special characters as it is.

Source Code:

```
#include<stdio.h>

void convertCase(char *str);

main()
{
    char str[100];
    clrscr();
    printf("Enter the String: ");
    gets(str);
    convertCase(str);
    printf("Converted string: %s", str);
    getch();
}

void convertCase(char *str)
{
    while (*str)
    {
        if (islower(*str))
        {
            *str = toupper(*str);
        }
    }
}
```

```
        else if (isupper(*str))
        {
            *str = tolower(*str);
        }
        str++;
    }
}
```

Output:

```
Enter the String: Sacred Heart College
Converted String: sACRED hEART cOLLEGE
```

Ex. Part A8: Write C program to search for number of occurrences of number in a list of numbers using one dimensional array and display its position.

Source Code:

```
#include <stdio.h>

void searchOccurrences(int arr[], int, int);

int main(){
    int arr[20];
    int size, i, target;
    clrscr();
    printf("Enter the size of the array: ");
    scanf("%d", &size);
    if (size <= 0 || size > 20) {
        printf("\nInvalid size!\n");
        return 1;
    }

    printf("Enter %d elements: ", size);
    for (i = 0; i < size; i++){
        scanf("%d", &arr[i]);
    }
    printf("Enter the search element: ");
    scanf("%d", &target);
    searchOccurrences(arr, size, target);
    getch();
}
```

```

        return 0;
    }

void searchOccurrences(int arr[], int size, int target){
    int count = 0,i;
    printf("Position(s) of %d: ", target);
    for (i = 0; i < size; i++)
        if (arr[i] == target)
        {
            count++;
            printf(" %d ", i+1);
        }

    if (count == 0)
        printf("Not found");
    else
        printf("\nTotal occurrences: %d", count);
}

```

Output:

```

Enter the size of the array: 5
Enter 5 elements: 10 22 31 22 47
Enter the search element: 22
Position(s) of 22: 2 4
Total occurrences: 2

```

Ex. Part B1: Write C program to find the largest and smallest elements with their position in one-dimensional array.

Source Code:

```
#include <stdio.h>

void main(){
    int n, i, arr[20];

    int max, min, max_pos=0, min_pos=0;

    clrscr();

    printf("Enter the number of elements in the array: ");

    scanf("%d", &n);

    printf("Enter %d elements:\n", n);

    for (i = 0; i < n; i++){
        printf("Element %d: ", i);

        scanf("%d", &arr[i]);

    }

    max = arr[0];
    min = arr[0];

    for (i = 1; i < n; i++) {
        if (arr[i] > max)
        {
            max = arr[i];
            max_pos = i;
        }

        if (arr[i] < min)
        {
```

```
        min = arr[i];  
        min_pos = i;  
    }  
}  
  
printf("\nLargest element is %d at position %d\n", max, max_pos);  
printf("Smallest element is %d at position %d\n", min, min_pos);  
}
```

Output:

```
Enter the number of elements in the array: 5  
Enter 5 elements:  
Element 0: 12  
Element 1: 42  
Element 2: 81  
Element 3: 35  
Element 4: 29
```

```
Largest element is 81 at position 2  
Smallest element is 12 at position 0
```

Ex. Part B2: Write C program to read N integer values into a single dimension array and arrange them in ascending order using bubble sort method.

Source Code:

```
#include <stdio.h>

void main() {

    int n, i, j, temp, arr[25];

    clrscr();

    printf("Enter the number of elements: ");

    scanf("%d", &n);


    printf("Enter %d elements:\n", n);

    for (i = 0; i < n; i++) {

        printf("Element %d: ", i);

        scanf("%d", &arr[i]);

    }


    for (i = 0; i < n - 1; i++)

        for (j = 0; j < n - i - 1; j++)

            if (arr[j] > arr[j + 1])

                {

                    temp = arr[j];

                    arr[j] = arr[j + 1];

                    arr[j + 1] = temp;

                }
```

```
printf("\nArray in ascending order:\n");  
  
for (i = 0; i < n; i++)  
    printf(" %d ", arr[i]);  
  
printf("\n");  
  
}
```

Output:

```
Enter the number of elements: 5  
Enter 5 elements:  
Element 0: 12  
Element 1: 36  
Element 2: 70  
Element 3: 43  
Element 4: 88  
  
Array in ascending order:  
12 36 43 70 88
```

Ex. Part B3: Menu driven program to perform addition, multiplication of two matrices.

Source Code:

```
#include<stdio.h>

int a[10][10],b[10][10];

int i,j,x,y;

void matrixadd();

void matrixmul();

void readmatrix();

void main(void)
{
    int ch;

    clrscr();

    do
    {

        printf("\nMatrix operations");
        printf("\n1.Addition");
        printf("\n2.Multiplication");
        printf("\n3.Exit");
        printf("\nEnter your choice:");
        scanf("%d",&ch);

        if(ch==1)

            matrixadd();

        else if(ch==2)

            matrixmul();
```

```

        else if(ch==3)
            exit(0);
        else
        {
            printf("Invalid Choice, Exiting!...\n");
            getch();
            exit(0);
        }
    }while(ch==1 ||ch==2 || ch==3);
    getch();
}

```

```

void readmatrix()
{
    printf("\nEnter order of the matrices a and b:");
    scanf("%d %d",&x,&y);
    printf("\nEnter matrix a :\n");
    for(i=0;i<x;i++)
        for(j=0;j<y;j++)
            scanf("%d",&a[i][j]);

    printf("\nEnter matrix b:\n");
    for(i=0;i<x;i++)
        for(j=0;j<y;j++)

```

```

scanf("%d",&b[i][j]);

}

void matrixadd()
{
    readmatrix();
    printf("\nMatrix a+b is:\n");
    for(i=0;i<x;i++)
    {
        for(j=0;j<y;j++)
            printf("%d\t",a[i][j]+b[i][j]);
        printf("\n");
    }
}

```

```

void matrixmul()
{
    readmatrix();
    printf("\n matrix a*b is :\n");
    for(i=0;i<x;i++)
    {
        for(j=0;j<y;j++)
            printf("%d\t",a[i][j]*b[i][j]);
    }
}

```

```
        printf("\n");  
    }  
}
```

Output:

Matrix operations

1.Addition

2.Multiplication

3.Exit

Enter your choice:1

Enter order of the matrices a and b:2 2

Enter matrix a :

5 2

7 9

Enter matrix b:

2 5

2 8

Matrix a+b is:

7 7

9 17

Matrix operations

1.Addition

2.Multiplication

3.Exit

Enter your choice:2

Enter order of the matrices a and b:2 2

Enter matrix a :

5 2

7 9

Enter matrix b:

2 5

2 8

Matrix a*b is :

10 10

14 72

Ex. Part B4: Write C program to find nCr and nPr using recursive function to calculate factorial.

Source Code:

```
#include<stdio.h>

#include<conio.h>

long factorial(int);

long find_ncr(int,int);

long find_npr(int,int);

int main()
{
    int n,r;
    long ncr,npr;
    clrscr();
    printf("\nEnter the value for n and r: ");
    scanf("%d %d",&n,&r);
    ncr=find_ncr(n,r);
    npr=find_npr(n,r);
    printf("\n%d c %d = %ld\n",n,r,ncr);
    printf("\n%d p %d = %ld\n",n,r,npr);
    getch();
    return 0;
}

long find_ncr(int a,int b)
{
```

```
        return(factorial(a)/(factorial(b)*factorial(a-b)));
    }
```

```
long find_npr(int a,int b)
{
    return(factorial(a)/factorial(a-b));
}
```

```
long factorial(int c)
{
    if(c==1||c==0)
        return 1;
    else
        return c*factorial(c-1);
}
```

Output:

Enter the value for n and r: 5 2

5 c 2 = 10

5 p 2 = 20

Ex. Part B5: Write C program to read a string and count numbers of letters, digits, vowels, constants, spaces and special characters present in it using user defined function.

Source Code:

```
#include<stdio.h>
#include<string.h>
#include<ctype.h>
void getdet(char str1[]);
void main() {
    char str[100];
    clrscr();
    printf("\nOUTPUT");
    printf("\nEnter a text:\n");
    gets(str); getdet(str);
}
void getdet(char str1[]){
int lcount=0,vcount=0,dcount=0,spcount=0,scount=0,cons=0,i,n;
char c;
n=strlen(str1);
for(i=0;i<n;i++){
if(isalpha(str1[i])>0)
{
lcount++;
c=toupper(str1[i]);
if(c=='A' || c=='E' || c=='I' || c=='O' || c=='U')
```

```

vcount++;

else

cons++;

}

else if(isdigit(str1[i])>0)

dcount++;

else if(isspace(str1[i])>0)

scount++;

else

spcount++;

}

    printf("\nThe number of letters in the text
is:%d\n",lcount);
    printf("\nThe number of vowels in the text
is:%d\n",vcount);
    printf("\nThe Number of consonants in the text is
%d\n",cons);
    printf("\nThe number of spacesin the text
is:%d\n",scount);
    printf("\nThe number of digits in the text
is:%d\n",dcount);
    printf("\nThe number of Special Charectors in the
text is:%d\n",spcount);
    printf("\n*****END*****");
    getch();

}

```

Output:

```
*****OUTPUT*****
Enter a text:
Sacred Heart College 574224

The number of letters in the text is:18

The number of vowels in the text is:7

The Number of consonants in the text is 11

The number of spacesin the text is:3

The number of digits in the text is:6

*****END*****
```

Ex. Part B6: Write C program to sort a list of strings in ascending order using pointers.

Source Code:

```
#include <stdio.h>

#include <string.h>

int main()
{
    char str[10][50], *ptr[10], *temp;
    int n, i, j;
    clrscr();
    printf("\nEnter the number of strings MAX 10. \n");
    scanf("%d",&n);
    getchar();
    printf("Enter the strings:\n");
    for (i = 0; i < n; i++){
        gets(str[i]);
        ptr[i] = str[i];
    }

    for (i = 0; i < n-1; i++)
        for (j = i + 1; j < n; j++){
            if (strcmp(ptr[i], ptr[j]) > 0){
                temp = ptr[i];
                ptr[i] = ptr[j];
                ptr[j] = temp;
            }
        }
}
```

```

        ptr[j] = temp;
    }

}

printf("\nSorted strings in ascending order:\n");

for (i = 0; i < n; i++)
    puts(ptr[i]);

return 0;
}

```

Output:

```

Enter the number of strings MAX 10.
5
Enter the strings:
Date
Cherry
Grape
Apple
Banana

Sorted strings in ascending order:
Apple
Banana
Cherry
Date
Grape

```

Ex. Part B7: Write a C program to enter the information of a student like name, register number, marks in three subjects into a structure and display total, average and grade, display details in a neat form.

Source Code:

```
#include <stdio.h>
#include <conio.h>

char getgrade(int);

void main(void)
{
    struct
    {
        char name[20];
        char regno[15];
        int mark[3];
    } student;

    char grade;
    int total;
    float average;

    clrscr();
    printf("Enter the register number: ");
    scanf("%s", student.regno);
    printf("Enter the name of the student: ");
```

```

scanf("%s", student.name);

printf("Enter the marks in FIT, COA, Programming in C (out of 100):\n");

scanf("%d %d %d", &student.mark[0], &student.mark[1],
&student.mark[2]);

total = student.mark[0] + student.mark[1] + student.mark[2];

average = total / 3.0;

if (student.mark[0] >= 35 && student.mark[1] >= 35 && student.mark[2]
>= 35)

    grade = getgrade(average);

else

    grade='F';

printf("\n===== STUDENT DETAILS =====\n");

printf("Name      : %s\n", student.name);

printf("Reg. No   : %s\n", student.regno);

printf("Marks     : %d, %d, %d\n", student.mark[0], student.mark[1],
student.mark[2]);

printf("Total      : %d\n", total);

printf("Average    : %.2f\n", average);

printf("Grade      : %c\n", grade);

printf("=====\n");

getch(); // wait for key press before exit

```

```
}
```

```
char getgrade(int avg)
```

```
{
```

```
    char r;
```

```
    if (avg >= 80)
```

```
        r = 'A';
```

```
    else if (avg >= 60)
```

```
        r = 'B';
```

```
    else
```

```
        r = 'C';
```

```
    return r;
```

```
}
```

Output:

```
Enter the register number: U05SH25S0002
```

```
Enter the name of the student: Ashwith
```

```
Enter the marks in FIT, COA, Programming in C (out of 100):  
90 95 82
```

```
===== STUDENT DETAILS =====
```

```
Name       : Ashwith  
Reg. No    : U05SH25S0002  
Marks      : 90, 95, 82  
Total      : 267  
Average    : 89.00  
Grade      : A
```

```
=====
```

Ex. Part B8: Write a menu driven program to,

A. Create a text file.

B. Append the contents of a text file to another existing file by accepting filenames.

C. Display the content of entered filename.

D. Exit.

Source Code:

```
#include<stdio.h>

#include<io.h>

void createfile();
void fileappend();
void dispfile();
void main(void)
{
    char ch;

    clrscr();

    do
    {
        printf("a. Create a Text File \n");
        printf("b. Append contents of text file to another existing file\n");
        printf("c. Display the content of entered File name \n");
        printf("d. Exit \n");
        scanf(" %c",&ch);
        if(ch=='a')
```

```

        createfile();
    else if(ch=='b')
        fileappend();
    else if(ch=='c')
        dispfile();
    else if(ch=='d')
        exit(0);
} while(ch=='a' || ch=='b' || ch=='c');
}

void createfile() {
    char file1[50]=" ";
    FILE *f1;
    clrscr();
    printf("Enter Textfile name to create:\n");
    scanf("%s",&file1);
    if((f1=fopen(file1,"r"))==NULL) {
        f1=fopen(file1,"w");
        fclose(f1);
        printf("\nText file created successfully\n");
        getch();
    } else
        printf("File already exists.....\n");
}

```

```

void fileappend() {
    char file1[256]="";
    char file2[256]="";
    FILE *f1, *f2;
    clrscr();
    printf("Enter Textfile name to copy:\n");
    scanf("%s",file1);
    if((f1=fopen(file1,"r"))==NULL) {
        printf("\nFile not found.....\n");
    } else{
        printf("\nEnter the existing File name to append:\n");
        scanf("%s",file2);
        if((f2=fopen(file2,"a"))==NULL)
            printf("File to be appended not found....\n");
        else{
            char ch;
            fseek(f2, 0, SEEK_END);
            while ((ch = fgetc(f1)) != EOF) {
                fputc(ch, f2);
            }
            printf("File copied successfully from '%s' to '%s'\n", file1,
file2);
            fclose(f1);

```

```

        fclose(f2);

    }

}

getch();

}

void dispfile() {

    FILE *fp;

    char c,file1[50];

    printf("Enter Textfile name to display:\n");

    scanf("%s",&file1);

    fp = fopen(file1, "r");

    if (fp == NULL)
        printf("Error: Could not open the file.\n");
    else
        while ((c = fgetc(fp)) != EOF) {
            putchar(c);
        }
    fclose(fp);
    printf("\n\n");

}

```

Output :

```

a. Create a Text File
b. Append contents of text file to another existing file
c. Display the content of entered File name
d. Exit
a

```

```

Enter Textfile name to create:
asdf.txt

```

```

Text file created successfully

```

- a. Create a Text File
 - b. Append contents of text file to another existing file
 - c. Display the content of entered File name
 - d. Exit
- b_

Enter Textfile name to copy:

abc.txt

Enter the existing File name to append:

xyz.txt

File copied successfully from 'abc.txt' to 'xyz.txt'

- a. Create a Text File
- b. Append contents of text file to another existing file
- c. Display the content of entered File name
- d. Exit

c

Enter Textfile name to display:

xyz.txt

this is a text file from c

Content of abc.txt

- a. Create a Text File
- b. Append contents of text file to another existing file
- c. Display the content of entered File name
- d. Exit